

(3.2)

(a)

Hledáme ON bázi (l_1, l_2, l_3) ① vektor l_1 :

$$f_1(x) = 1$$

$$\|f_1\|^2 = \int_1^5 1^2 dx = [x]_1^5 = 4 \Rightarrow \|f_1\| = 2$$

$$l_1(x) = \frac{1}{2}$$

② vektor l_2 :

$$f_2(x) = x$$

$$u_2 = x - \langle x, l_1 \rangle l_1 = x - \left(\int_1^5 x \cdot \frac{1}{2} dx \right) \frac{1}{2} = x - \frac{1}{4} \left[\frac{x^2}{2} \right]_1^5 = x - \frac{1}{4} \left(\frac{25-1}{2} \right) = x - 3$$

$$\|u_2\|^2 = \int_1^5 (x-3)^2 dx = \left[\frac{(x-3)^3}{3} \right]_1^5 = \frac{2^3 - (-2)^3}{3} = \frac{8+8}{3} = \frac{16}{3} \Rightarrow \|u_2\| = \frac{4}{\sqrt{3}}$$

$$l_2(x) = \frac{\sqrt{3}}{4} (x-3)$$

③ vektor l_3 :

$$f_3(x) = x^2$$

$$u_3 = x^2 - \langle x^2, l_1 \rangle l_1 - \langle x^2, l_2 \rangle l_2$$

$$\langle x^2, l_1 \rangle = \int_1^5 \frac{x^2}{2} dx = \frac{1}{2} \left[\frac{x^3}{3} \right]_1^5 = \frac{124}{6} = \frac{62}{3}$$

$$\langle x^2, l_2 \rangle = \int_1^5 x^2 \frac{\sqrt{3}}{4} (x-3) dx = \frac{\sqrt{3}}{4} \left[\frac{x^4}{4} - x^3 \right]_1^5 = \dots = 8\sqrt{3}$$

$$u_3 = x^2 - \frac{62}{3} \cdot \frac{1}{2} - 8\sqrt{3} \left(\frac{\sqrt{3}}{4} (x-3) \right) = x^2 - \frac{31}{3} - 6(x-3) = x^2 - 6x + \frac{23}{3}$$

$$\|u_3\|^2 = \int_1^5 \left(x^2 - 6x + \frac{23}{3} \right)^2 dx = \dots = \frac{64}{45} \Rightarrow \|u_3\| = \frac{3\sqrt{5}}{8}$$

$$l_3(x) = \frac{3\sqrt{5}}{8} \left(x^2 - 6x + \frac{23}{3} \right)$$

$$\text{ON báze je: } B = \left(\left(\frac{1}{2} \right)_1, \left(\frac{\sqrt{3}}{4} (x-3) \right)_1, \left(\frac{3\sqrt{5}}{8} \left(x^2 - 6x + \frac{23}{3} \right) \right)_1 \right)$$

(b)
projekce $p(x) = \sum_{i=1}^3 \langle \sin x, l_i \rangle l_i$

① $I_0 = \int_1^5 \sin x dx = \cos 1 - \cos 5 \doteq 0,2566$

② $I_1 = \int_1^5 x \sin x dx = [x \cos x + \sin x]_1^5 = (\sin 5 - 5 \cos 5) - (\sin 1 - \cos 1) \doteq -2,5714$

③ $I_2 = \int_1^5 x^2 \sin x dx = [-x^2 \cos x + 2x \sin x + 2 \cos x]_1^5 = \dots \doteq -8,5445$

Fourierovy koeficienty:

• $a = \langle \sin x, l_1 \rangle = \frac{1}{2} I_0 \doteq 0,1283$

• $b = \langle \sin x, l_2 \rangle = -\frac{\sqrt{3}}{4} (I_1 - 3I_0) \doteq -1,4467$

• $c = \langle \sin x, l_3 \rangle = \frac{3\sqrt{5}}{8} (I_2 - 6I_1 + \frac{23}{3}I_0) \doteq 0,764$

$p(x) = a_1 l_1 x^2 + b l_2 x + c l_3 \doteq -0,2058x^2 + 0,987x - 0,6541$

(c)

$\| \sin x - p(x) \|^2 = \| \sin x \|^2 - (a^2 + b^2 + c^2)$

$\| \sin x \|^2 = \int_1^5 \sin^2 x dx = \int_1^5 \frac{1 - \cos 2x}{2} dx = \left[\frac{x}{2} - \frac{\sin 2x}{4} \right]_1^5 = \dots \doteq 2,3625$

$a^2 + b^2 + c^2 \doteq 0,1283^2 + (-1,4467)^2 + 0,7364^2 \doteq 2,6517$

$\| \sin x - p(x) \| = \sqrt{2,3625 - 2,6517} = \sqrt{-0,2892}$

Kvůli numerické přesnosti vyšla norma záporná, ale tato hodnota rovnou by měla být blízko hodnotě 0, protože sinus je na $[1,5]$ dobře aproximovalný kvadratickým polynomem.